

How many capacitances is a device?

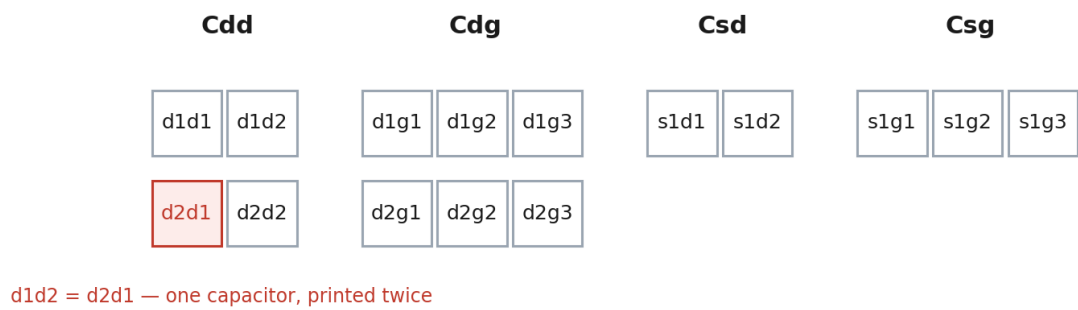
Four numbers are floating around — 15, 14, 13 and 9. This is where each of them comes from, and which one belongs in the paper.

The short answer

A device is **14 free parameters**. Figure 1(b) prints **15** numbers, but two of them are the same capacitor, so only 14 of them can be chosen independently.

1. Why 15 and not 14 — C_dd is symmetric

The mutual capacitance between dot 1 and dot 2 is one physical quantity. Writing it as a matrix puts it in twice, once above the diagonal and once below:



The four matrices as Figure 1(b) prints them. The shaded cell is not a separate capacitance — it is the same number as the one above the diagonal.

So counting the printed cells gives 15, and counting the independent values gives 14. The code enforces this rather than assuming it: MatrixGenerator.generate_all builds C_dd with generate_symmetric(), which draws the upper triangle and mirrors it.

Checked on a stored device rather than argued:
data/_device_pools/devices_n550_res100_c1df7b6bf/sample_100/device.json prints 15 numbers, 14 of them distinct, and its C_dd[0][1] equals C_dd[1][0].

matrix	shape	cells printed	free parameters
C_dd	2 x 2 (symmetric)	4	3
C_dg	2 x 3 (QArray's Cgd)	6	6
C_sd	1 x 2 (QArray's Cds)	2	2
C_sg	1 x 3 (QArray's Cgs)	3	3
total		15	14

2. Where 13 came from — it was simply wrong

13 was not a different way of counting. It was a number typed into the text and never checked against the code: the manuscript said a device is “a point in a 13-dimensional box” while, two paragraphs later, the separation sentence printed “14-dimensional normalised parameter space” from the data. Table I has always listed 14 rows. The text contradicted itself and the table.

It is fixed in three places, and neither number is typed any more — both are computed from the shapes the generator uses and from `DEFAULT_INTERVALS`, and the build asserts they agree.

3. Where 9 comes from — the drawing is incomplete

This is the one still worth acting on. Panel (a) of `docs/figures/dqd_model.png` draws 9 capacitors. Five that panel (b) tabulates are not in the drawing at all.

	Vg1	Vg2	Vg3	QD1	QD2	ground
QD1	d1g1	d1g2	d1g3		d1d2	d1d1 not drawn
QD2	d2g1	d2g2	d2g3 not drawn	d1d2 same as above		d2d2 not drawn
QDs	s1g1 not drawn	s1g2 not drawn	s1g3	s1d1	s1d2	

☐ drawn in Figure 1(a) (9) ☐ in the matrices, missing from Figure 1(a) (5)

Every capacitance the matrices define, as one cell: what it couples, and whether Figure 1(a) draws it. White: the 9 it draws. Red: the 5 it does not — the two dot self-capacitances to ground, the dot-2 coupling to gate 3, and the sensor's couplings to gates 1 and 2. The shaded QD2/QD1 cell is the symmetric duplicate, not a fifteenth capacitance.

Missing from panel (a): d1d1, d2d2, d2g3, s1g1, s1g2.

9 drawn plus 5 missing is 14, which is the same count arrived at from the matrices — so nothing is wrong with the physics or the code. What is wrong is that a referee who counts the picture will find 9 where the text now says 14.

4. What to do

- **Either** add the five missing capacitors to panel (a) in Illustrator, so (a) and (b) show the same network;
- **or** leave the drawing and say so in the caption — that (a) shows the couplings which set the geometry of the diagram and (b) the complete set of 14. Say which and I will write the caption.

The text in Section 3.1 is already correct either way: it states the symmetry, the 15 printed entries, the 14 free parameters and the per-matrix breakdown, all computed at build time.